A collaborative Web-Editor for Editing and Simulating Renew Reference Nets (Master's Thesis)

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Introduction and Motivation

- Petri nets are a family of graphical modeling languages used for the specification and verification of concurrent processes and dynamic systems [2].
- Renew is a tool for the design and simulation of Petri nets and has been under continuous development at the University of Hamburg since 1999 [1].
- However, Renew cannot be used on mobile devices and does not support real-time collaborative editing by multiple users.
- **In this work** we present a web-based Petri net editor that enables collaborative editing and simulation on desktop and mobile devices, backed by the Renew simulator.

Basics: Petri nets

- A net structure consists of passive elements (**places**; circular) and active elements (**transitions**; rectangular), which are connected by arrows (**arcs**).
- Places represent system states and containers for resources (**tokens**).
- Arcs and transitions represent actions and define the rules by which tokens may move from one place to another.

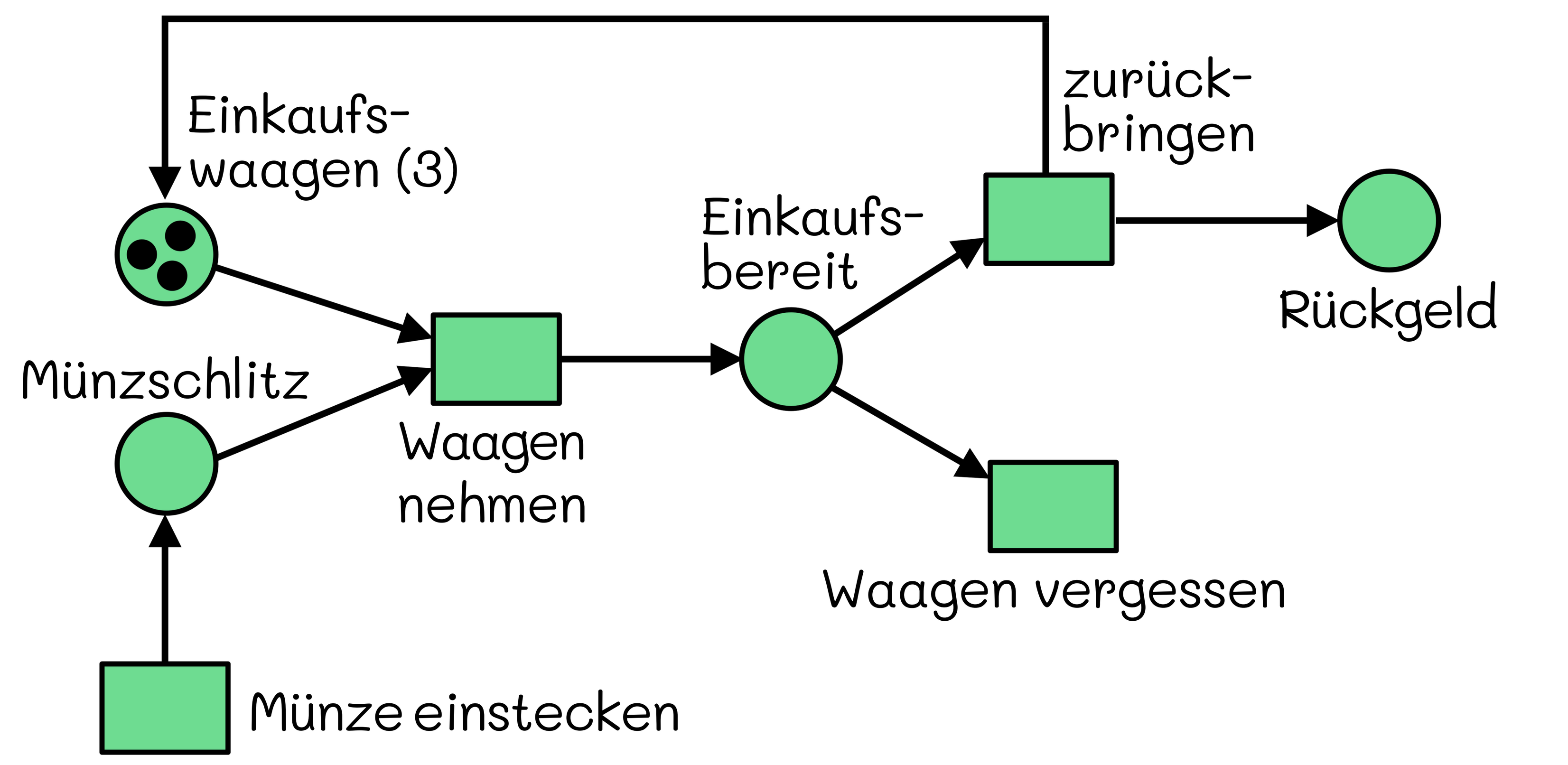

Figure 1: Example of a Petri net consisting of 4 places, 4 transitions, and 3 tokens in one place, modeling a shopping cart rental system

Requirements

- Existing Renew files can be imported and exported.
- Documents support realtime collaborative editing.
- The editor runs in a web browser on various devices (desktops, smartphones, tablets).
- Complex Petri nets can be simulated using the powerful Renew simulation engine.
- Interactive collaborative simulations are supported.

Implementation (System Architecture)

- Client/Server architecture, communication via HTTP and WebSocket
- Headless instances of Renew are hosted on the central server for running the simulation.

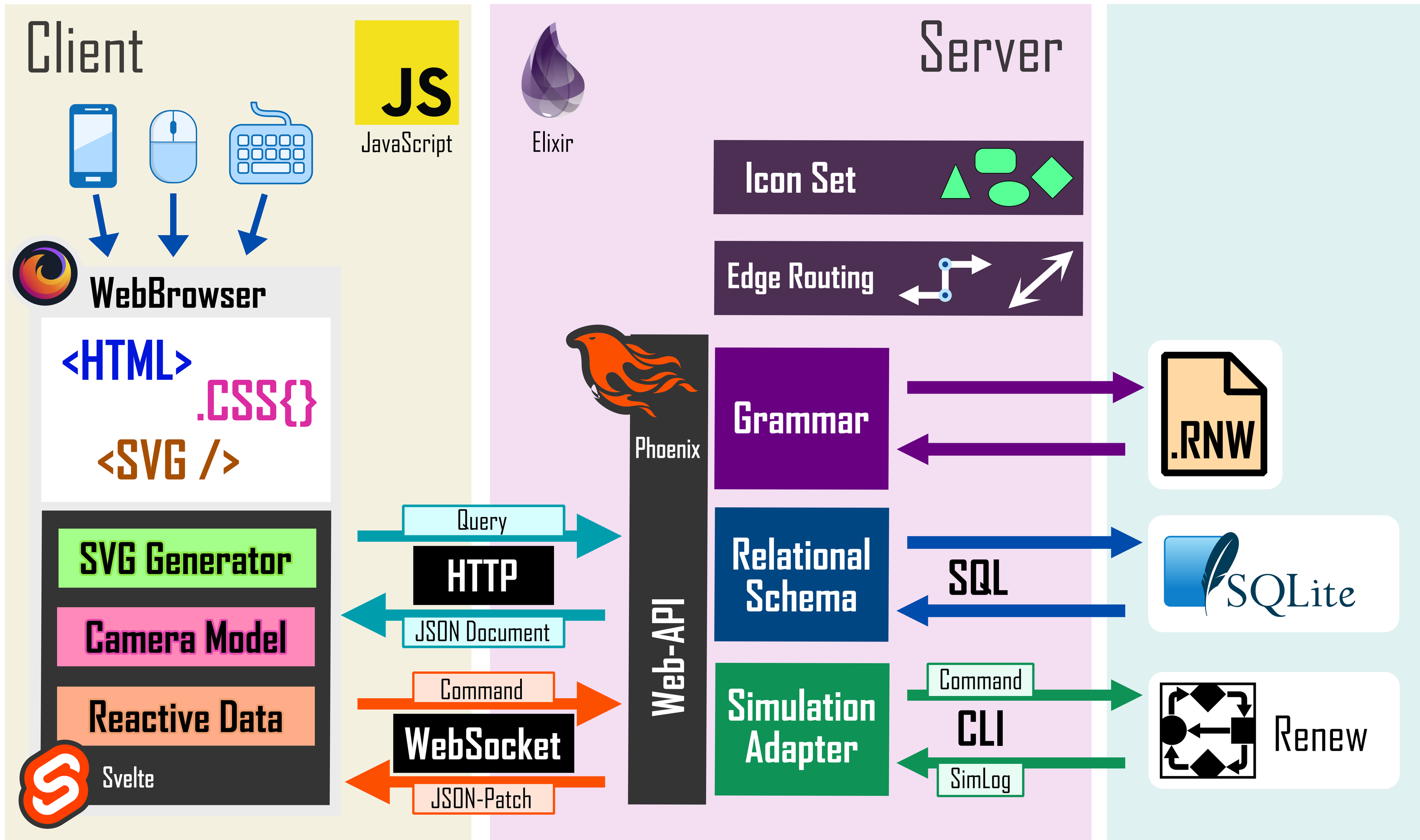

Figure 2: System architecture of the editor consisting of client and server components, with integration of Renew and a relational database

Result (User Interface)

- The interactive canvas provides various tools for manipulating and navigating Petri nets.

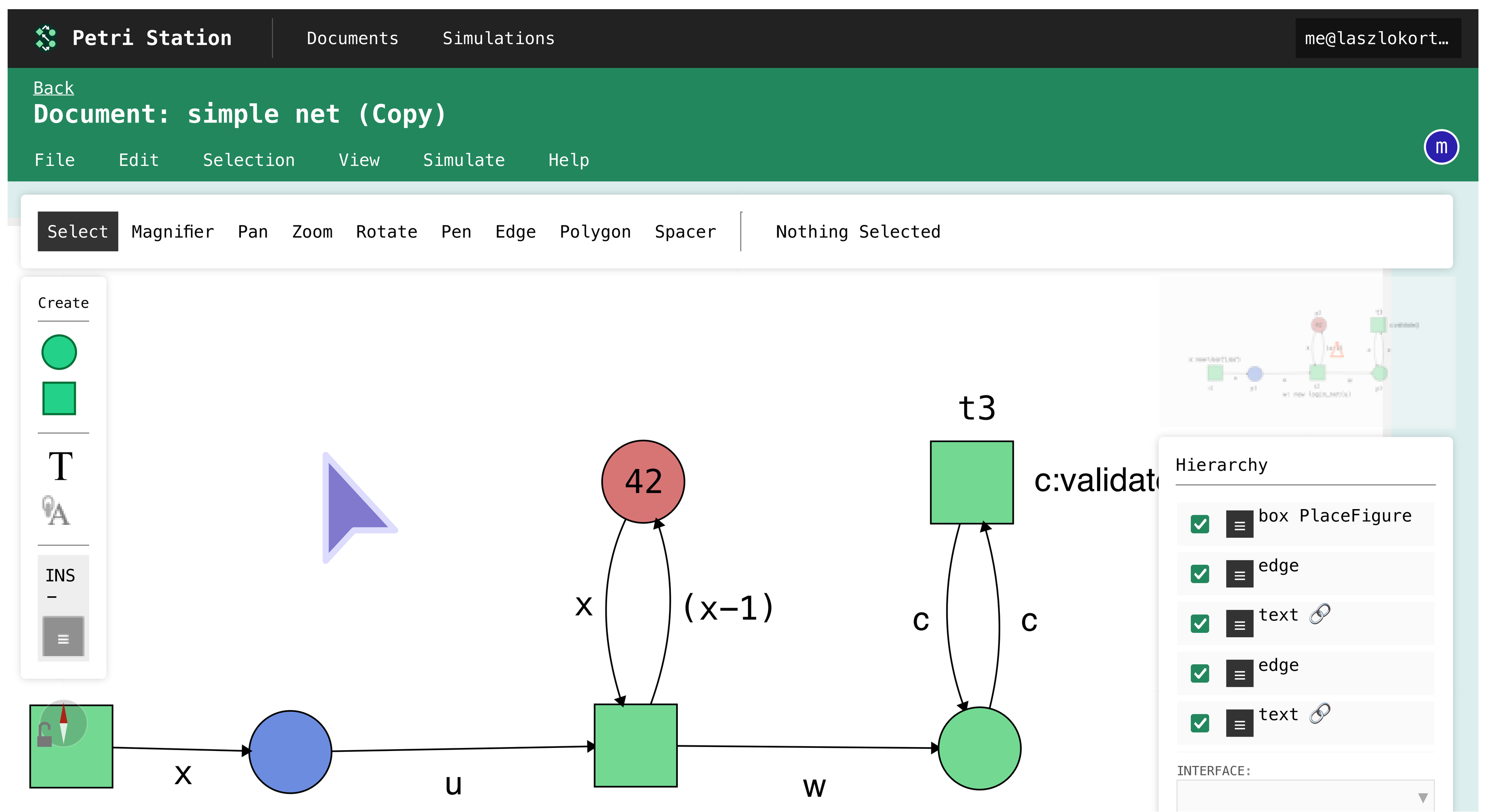

Figure 3: Screenshot of the editor user interface showing the toolbar (top), tool palette (left), mini-map(top right), element hierarchy (bottom right), and another user's collaborative cursor (center-left).

Try it yourself!

Scan the QR code to access the web editor.

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https://editor.petristation.net
Login:      demo@tgi
Password:   petri2025
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Outlook

A follow-up Bachelor's thesis with significant improvements is currently in progress.

References

[1] Olaf Kummer et al. *Renew – The Reference Net Workshop*. Available at: http://www.renew.de/. Release 1.0. Mar. 1999. URL: http://www.renew.de/.

[2] Carl Adam Petri. "General Net Theory". In: *Computing System Design: Proc. of the Joint IBM University of Newcastle upon Tyne Seminar, Sep., 1976*. Ed. by B. Shaw. University of Newcastle upon Tyne, 1977, pp. 131–169.